MILITARY SPECIFICATION

DETECTOR, WIND DIRECTION, SPEED, AND TEMPERATURE, FIRE AND FLIGHT AIR DATA SUBSYSTEM, HELICOPTER ARAMAMENT: M143

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification contains performance and environmental requirements, quality assurance provisions, and packaging for delivery for the Detector, Wind Direction, Speed, and Temperature, Fire and Flight Air Data Subsystem, Helicopter Armament: M143.

2. APPLICABLE DOCUMENTS.

2.1 Government documents. Unless otherwise specified, the following Specifications and Standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-E-6051	Electromagnetic Compatibility Requirements Systems
MIL-I-8500	Interchangeability and Replaceability of Components Parts for Aircraft and Missiles

FSC 1270

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-F-13926	Fire Control Materials: General Specification Governing the Manufacture and Inspection of
MIL-M-38510	Microcircuits, General Specification for
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication and Radar for Aerospace and Control Systems, Procedure for
MIL-S-46844	Solder Bath, Soldering and Printed Wiring Assemblies, Automatic Machine Type

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristcs and Utilization of
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810C	Environmental Test Methods

2.1.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified hrein.

DRAWINGS

US ARMY ARMAMENT, RESEARCH AND DEVELOPMENT CENTER (ARDC)

P11776800	Preservation and Packaging Data Sheet
11776800	Detector, Wind Direction Speed and Temperature
11776801	Tail Assembly
11776813	Neck Assembly
11776858	Cover and Dowel Assembly
11776864	Body and Plug Assembly
11776869	Plate and Connector Assembly
11776877	Head Assembly

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein:

American Society for Testing and Materials (ASTM)
ASTM-D3951 - Standard Practice for Commercial Packaging.

2.3 Order of precedence. In the event of a conflict of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications, and standards.

3.2 Parts. All parts shall comply with the applicable drawings, specifications, and standards.

3.3 Air Data System Airspeed and Direction Sensor (AADS). The AADS shall comply with all requirements specified on Drawing (DWG) 11776800, all associated drawings, and with all requirements specified in applicable specifications and standards. The AADS shall consist of a pitotstatic pressure head supported on a gimbal arrangement, caused to point into the local airflow by a finned tail. Total and static pressures shall be sensed by the head which is aligned with the local resultant airflow, and shall not suffer from incidence errors. Air pressure shall be conveyed via the axles of the gimbal to aircraft plumbing. Movement of the gimbal shall position two angular resolvers to generate electrical signals which represent the movement of the pitot head in pitch and yaw axes relative to the aircraft. These signals, together with the pressure signals, shall be used by the ADS EPU to compute helicopter airspeed and direction. A gimbal arrangement shall permit total angular freedom in pitch and $\pm 60^{\circ}$ in yaw to allow the head to sense accurately the local pitot and static pressures for helicopter foward and lateral airspeeds. The pressure head shall be equipped with a heating element.

The AADS shall incorporate an Air Temperature Sensor (ATS) which does not interfere with airflow over the pressure head or pick up heated air from the head.

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3.3.1 Major component list.

Nomenclature	Drawing
Head Assembly	11776877
Plate and Connector Assembly	11776869
Body and Plug Assembly	11776864
Cover and Dowel Assembly	11776858
Neck Assembly	11776813
Tail Assembly	11776801

3.4 Characteristics.

3.4.1 Electrical.

3.4.1.1 Signal interface. The AADS signal interface with the EPU shall be as defined in Appendix A to this specification and drawing 11776300.

3.4.1.2 Electrical power.

3.4.1.2.1 Steady state. The electrical service conditions will be in accordance with MIL-STD-704, Category "B" utilization equipment, except that the transient frequency limit of Figure 5, limit 4 in MIL-STD-704 shall be 240Hz. The equipment shall meet all specified requirements of MIL-STD-704 for the types of power specified herein, and shall give specified performance when supplied with electrical power having these characteristics and limits. The power required shall not exceed the amounts specified below:

a. 115/200 Volts, 400Hz, 1-phase grounded neutral 200 Volt amperes (anti-icing).

3.4.1.2.2 Electrical power interruptions. Momentary power interruptions to the equipment will be experienced when external primary power is switched to helicopter power or when switching from one helicopter power bus to another. The ADS shall meet the performance requirements of this specification after such momentary power interruptions have been experienced. The ADS shall return automatically to specified performance within 5 seconds after any power interruption.

3.4.1.3 Mechanical interface. The maximum envelope dimensions and mounting provisions of the AADS shall conform to drawing 11776806.

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3.4.1.4 Interfacing components. The AADS shall interface with the ADS EPU and pitot head heater control and pitot static plumbing on drawing 11776300.

3.4.2 Performance.

3.4.2.1 Input/output parameter range. The AADS shall have input/output characteristics over the range and within the tolerances specified in Table 1.

3.4.2.2 Accuracy. The AADS accuracy values shown in Table I shall be defined as 95 percent confidence values. AADS measured accuracy shall be within the specified accuracies in at least 19 out of each 20 separate measurements within the ADS performance envelope specified in Table I.

3.4.2.3 Anti-icing provisions. The AADS shall contain an anti-icing subsystem. The anti-icing subsystem shall use 115 Vac, 400Hz, single phase helicopter power for the heating function.

3.4.2.4 Mechanical alignment. The AADS shall have a reference plane or mounting points with respect to which the sensor shall be tested for directional airspeed accuracy. Mounting provisions shall be in accordance with 11776806.

3.4.2.5 Orthogonality. AADS pitch and yaw electrical outputs shall be such that the equivalent measurement axes are mutually perpendicular within 1 degree.

3.4.2.6 System turn-on time. The ADDS response time from power-on actuation to ready state shall not exceed 5 seconds.

3.4.2.7 Air leakage. Air leakage within the AADS from the pitot and static lines shall not cause an equivalent error of greater than 0.5 knot in the dynamic pressure sensed by the AADS. This shall apply for any point within performance envelope of the ADS, and for the environmental conditions specified herein.

3.4.2.8 Freedom of movement. The pitot/static sensing assembly shall have adequate freedom of movement to accurately sense the parameters listed in Table 1.

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3.4.3 Physical characteristics.

3.4.3.1 Dimensions. Dimensional requirements applicable to envelop size, mounting and mating shall be in accordance with drawing 11776806.

3.4.3.2 Weight. The maximum weight of the AADS shall be 2 pounds 7 ounces.

3.4.4 Precondition burn in. Each AADS procured under this specification shall be subjected to a burn-in test as defined in Figure 1.

3.4.5 Environment conditions. The AADS shall operate and meet the performance requirements of 3.4.2 under any of the environmental conditions, or a reasonable combination of these conditions, as specified herein.

TABLE I. Input/output parameter range and characteristics

Input/ Output	Parameter	Units	Nom. Value/ Range	Tolerance (Plus or Minus)
Output	Air Temperature Sensor	Ohms	367 thru 617	2 ohm/^{o}C
Input	Pitot Head Heater	400 Hz Volts AC	115.0	5.0
Input	Resolver Supply	Volts AC	19.0	2.0
Output	Angle (Pitch) (α)	degrees	0 thru 360	2.0
Output	Angle (Pitch-Yaw) (α-B)	degrees	0 thru 360	2.0

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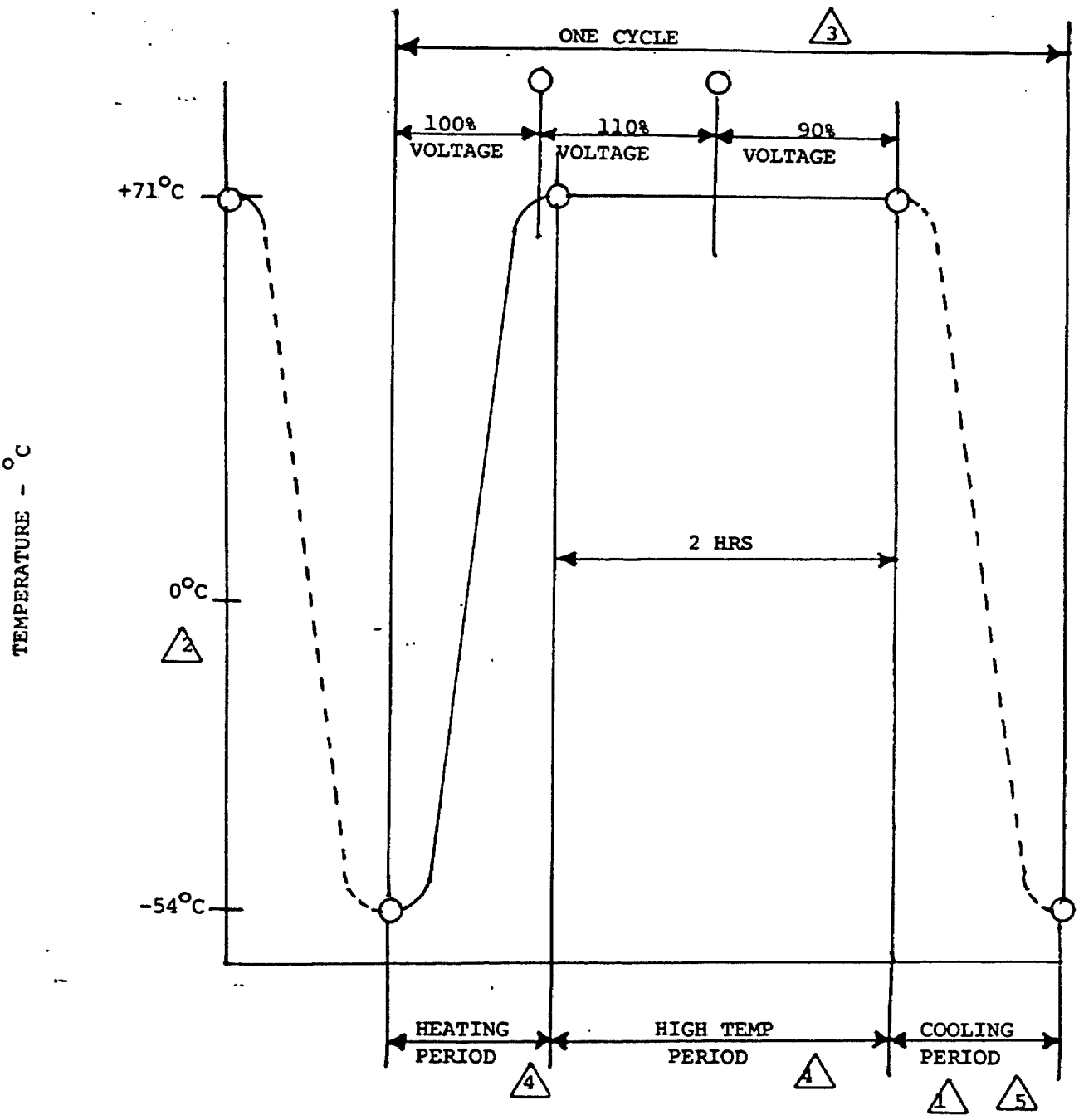


FIGURE 1. Typical temperature cycle - electronic and avionic equipment - continued

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NOTES:

1. During the cooling period, the chamber temperature shall not be lower than -60°C.

2. Once every 24 hours at 0°C, moisture shall be injected until the equipment being tested is coated with frost.

3. One temperature cycle shall start with the equipment stabilized at the low temperature, heated and stabilized at the high temperature, maintained at the high temperature for 2 hours, then end with the equipment cooled and stabilized at the low temperature. During the temperature cycle, the equipment shall be operated in accordance with its normal duty cycle.

4. Equipment ON; input voltage at 110% and 90%, respectively during each 1/3 of the total time of heating period plus high temperature period; momentarily OFF between input voltage changes.

5. Equipment OFF.

6. Vibration -2.2g ±10% peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The vibration shall be at least 10 minutes during each hour of operating time.

7. Transition rate between temperature levels shall average 5°C per minute.

FIGURE 1. Typical temperature cycle - electronic and avionic equipment

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3.4.5.1 Natural environment. The AADS shall operate and withstand storage in the environment described in MIL-E-5400 except as noted herein.

3.4.5.1.1 Altitude. Operating, 0-20,000 feet. Nonoperating and storage 0-50,000 feet.

3.4.5.1.2 Temperature. Operating, nonoperating, and storage -54°C to +71°C,

3.4.5.1.3 Humidity. Operation while being subjected to MIL-STD-810, Method 507.1, Procedure II. In Step 6, measurements shall be taken during the last five hours of the last test cycle.

3.4.5.1.4 Sand and Dust. Operation after being subjected to MIL-STD-810, Method 510.1, Procedure I, except that the air velocity in Steps 1, 2, and 3 shall be 200 ± 100 fpm. Step 3 may proceed immediately after stabilization in Step 2, if desired.

3.4.5.1.5 Salt atmosphere. Operation after being subjected to MIL-STD-810, Method 509.1. Operation of the AADS shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

3.4.5.1.6 Fungus. No evidence of viable fungus on any surfaces after being subjected to MIL-STD-810, Method 508.1.

3.4.5.1.7 Explosive atmosphere. Operation after being subjected to MIL-STD-810, Method 511.1, Procedure I, except test to be limited to a maximum altitude of 20,000 feet.

3.4.5.1.8 Rain. Operation after being subjected to MIL-STD-810, Method 506.1, Procedure I. During each thirty minute rain period, the wind shall be maintained for fifteen minutes, commencing five minutes after the start of the period as specified in the procedures. The AADS shall be operated during the last ten minutes of the thirty minute period.

3.4.5.2 Service environment. The AADS shall withstand the service environment as specified in MIL-E-5400 for Class 2 equipment, except as noted in paragraphs 3.4.5.1.1 thru 3.4.5.1.8.

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3.4.5.2.1 Bench handling. The AADS shall withstand shock environments encountered in maintenance and servicing.

3.4.5.3 Induced environment.

3.4.5.3.1 Vibration. The AADS shall not sustain damage or subsequently fail to provide specified performance when subjected to the vibration environment defined by Figure 2, except that the maximum double amplitude shall be 0.20 inch.

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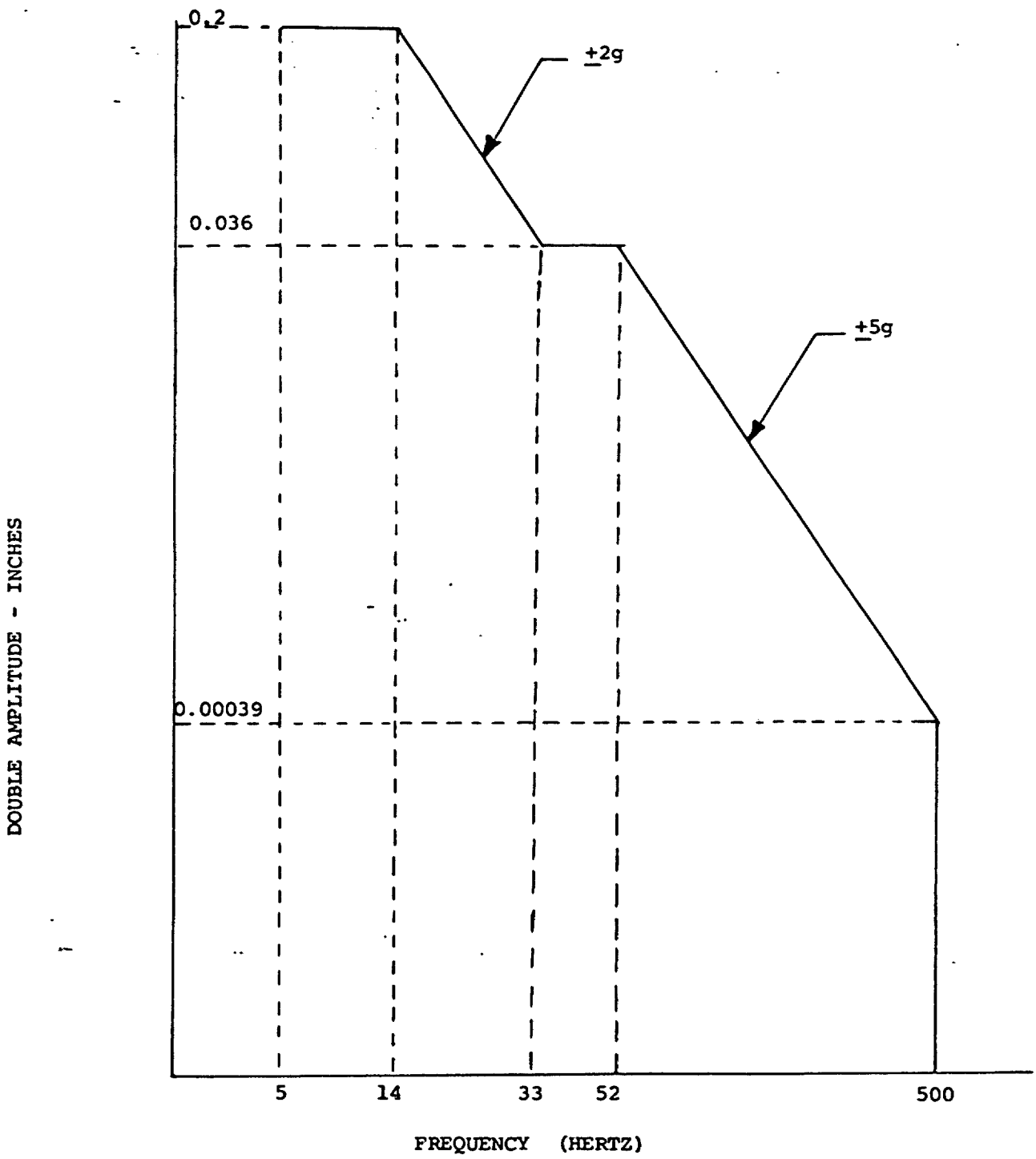


FIGURE 2. Vibration amplitude for qualification test

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3.4.5.3.2 Shock. The AADS shall meet the shock requirements of MIL-E-5400. The requirements of MIL-E-5400 for mounting hardware crash safety shall apply only to the units mounted inside the cockpit.

3.4.6 Transportability. The AADS shall withstand vibration environments encountered during transportation.

3.4.7 Processes.

3.4.7.1 Soldering. All soldering shall be performed in accordance with MIL-S-45743 or MIL-S-46844.

3.4.7.2 Microelectronic devices. Microelectronic devices shall conform to MIL-M-38510.

3.4.8 Electromagnetic radiation.

3.4.8.1 Electromagnetic compatibility. Each AADS installed in the helicopter as part of the ADS shall not cause undesirable response, malfunction, or degraded performance to any other equipment/subsystem installed in or associated with the helicopter; nor shall any component malfunction, exhibit degraded performance, or show undesirable response beyond the tolerances specified in MIL-E-6051 when other equipment on board the helicopter are singularly or collectively operated and performing their intended functions.

3.4.8.2 Electromagnetic interference. The AADS when operated as part of the ADS shall meet the requirements of MIL-STD-461 equipment when tested to the following requirements: CE01, CE03, CE04, CE05, CS01, CS02, CS06, RE02, RS01, RS03, RS03.1, and RS02.

3.4.8.3 Circuit isolation. Signal and power potentials shall be electrically isolated from each other and the equipment chassis.

3.4.9 Interchangeability. All components having the same manufacturer's part number shall be dimensionally and functionally interchangeable in accordance with MIL-I-8500. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part number and changes thereto. Interchangeability must be maintained within the detail parts of the level of replaceable items.

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3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship. Workmanship shall be in accordance with MIL-F-13926, electronic and mechanical weapon-build practices, to ensure safety, proper and dependable operation, and service life. Particular attention shall be given to assure thoroughness of assembly, alignment of parts, electrical connections, cleanliness, tightness of screws and bolts, marking of parts, application of protective finish, and freedom from burrs and rough or sharp edges.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classed as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926.

4.4. First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three AADS selected at random by the Government representative from the first fifteen production AADS.

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4.4.2 Inspection. The sample shall be subjected to all the inspections specified in Table II and III.

4.4.3 Failure. Failure of any AADS to meet the requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any AADS to comply with any stated requirement.

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TABLE II. First article tests

Item	Characteristic	Requirement	Test Procedure
1	Natural environment	3.4.5.1	4.6.4.1
2	Altitude	3.4.5.1.1	4.6.4.2
3	Temperature	3.4.5.1.2	4.6.4.3
4	Humidity	3.4.5.1.3	4.6.4.4
5	Sand and dust	3.4.5.1.4	4.6.4.5
6	Salt atmosphere	3.4.5.1.5	4.6.4.6
7	Fungus	3.4.5.1.6	4.6.4.7
8	Explosive atmosphere	3.4.5.1.7	4.6.4.8
9	Rain	3.4.5.1.8	4.6.4.9
10	Interchangeability	3.4.9	Applicable drawings,visual, MIL-I-8500 and MIL-STD-100
11	Service environment	3.4.5.2	4.6.4.10
12	Bench handling	3.4.5.2.1	4.6.4.10.1
13	Induced environment	3.4.5.3	4.6.4.11
14	Vibration	3.4.5.3.1	4.6.4.11.1
15	Shock	3.4.5.3.2	4.6.4.11.2
16	Electromagnetic radiation	3.4.8	4.6.4.12
17	Transportability	3.4.6	4.6.4.11.3

TABLE III. First article tests

Item	Characteristic	Requirement	Test Procedure
101	Air temperature	3.4.2.2	4.6.2.9
102	Pitch angle (α)	3.4.2.2	4.6.2.10
103	Pitch-Yaw Angle ($\alpha - \beta$)	3.4.2.2	4.6.2.10
104	Air leakage	3.4.2.7	4.6.2.5

TABLE IV. General tests

Item	Characteristic	Requirement	Test Procedure
301	Fabrication and General Specifications	3.1., 3.2 & 3.3	Applicable drawings,visual and MIL-E-5420 visual

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4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

4.5 Quality conformance inspection.

4.5.1 Performance inspection.

4.5.1.1 Sample and tests. Each AADS in every lot shall be subjected to the tests in Table III and shall be examined visually for completeness, improper assembly, and evidence of poor workmanship.

4.5.1.2 Rejection. Where any AADS fails to meet any specified requirement, the defective AADS shall be rejected and removed from the lot.

4.5.2 General inspection.

4.5.2.1 Sample and tests. One AADS, as a control sample, shall be selected at random by the Government representative from each 100 AADS produced and shall be subjected to the tests in Table IV.

4.5.2.2 Rejection. Where any AADS fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.6 Methods of inspection.

4.6.1 Test equipment. The test equipment in Table V shall be used where specified to perform the required tests.

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TABLE V. Test equipment

ITEM/NOMENCLATURE	TYPE		MFR, MODEL OR PART NO
	STANDARD	PECULIAR	
1. Test Set ADS (LRU) (M142)		X	11779002
2. Digital Voltmeter	X		Fluke 8800 or equivalent
3. Multimeter	X		Avometer Model 8 or equivalent
4. AADS Pneumatic Test Connector		X	11779070
5. Insulation Resistance Tester	X		Hewlett Packard HP429A or equivalent
6. Gram Gage	X		Capable of measuring 0-25 grams
7. Storage Tank	X		Capable of 1ft^3 Volume
8. Precision Pressure Gage	X		Texas Instruments Type 45 or equivalent
9. Thermometer	X		10° to 30°C
10. Checking Fixture		X	11779073
11. Weighing Machine	X		Capable of 0 to 2 pounds 80Z
12. Supply of clear dry air or nitrogen pressure regulated over the range zero to 255 knots			

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4.6.2 Performance tests. Power required by the AADS and the Test Set ADS (LRU) when under test shall be within the steady state limts of MIL-STD-704 and shall be less than:

28 Volts	DC	2 Amps
115 Volts	400 Hz	0.5 Amps

4.6.2.1 Torque test-yaw axis. Mount the AADS in the Fixture AADS (Checking) as shown in Figure 3. Using the gram gage, check that the torque required to cause rotation in the Yaw axis in both positive and negative directions is less than 100g-cm.

4.6.2.2 Torque test-pitch axis. Using the gram gage, check that the torque required to cause rotation in the Pitch axis in both positive and negative directions is less than 100g-cm.

4.6.2.3 Static pressure leak test. With the AADS connected as shown in Figure 4, raise the pressure to simulate an airspeed of 255 knots, and isolate the supply by closing valve A. When the airspeed has fallen to 250 knots, start the stopwatch. After 110 secs, the airspeed shall not be less than 50 knots.

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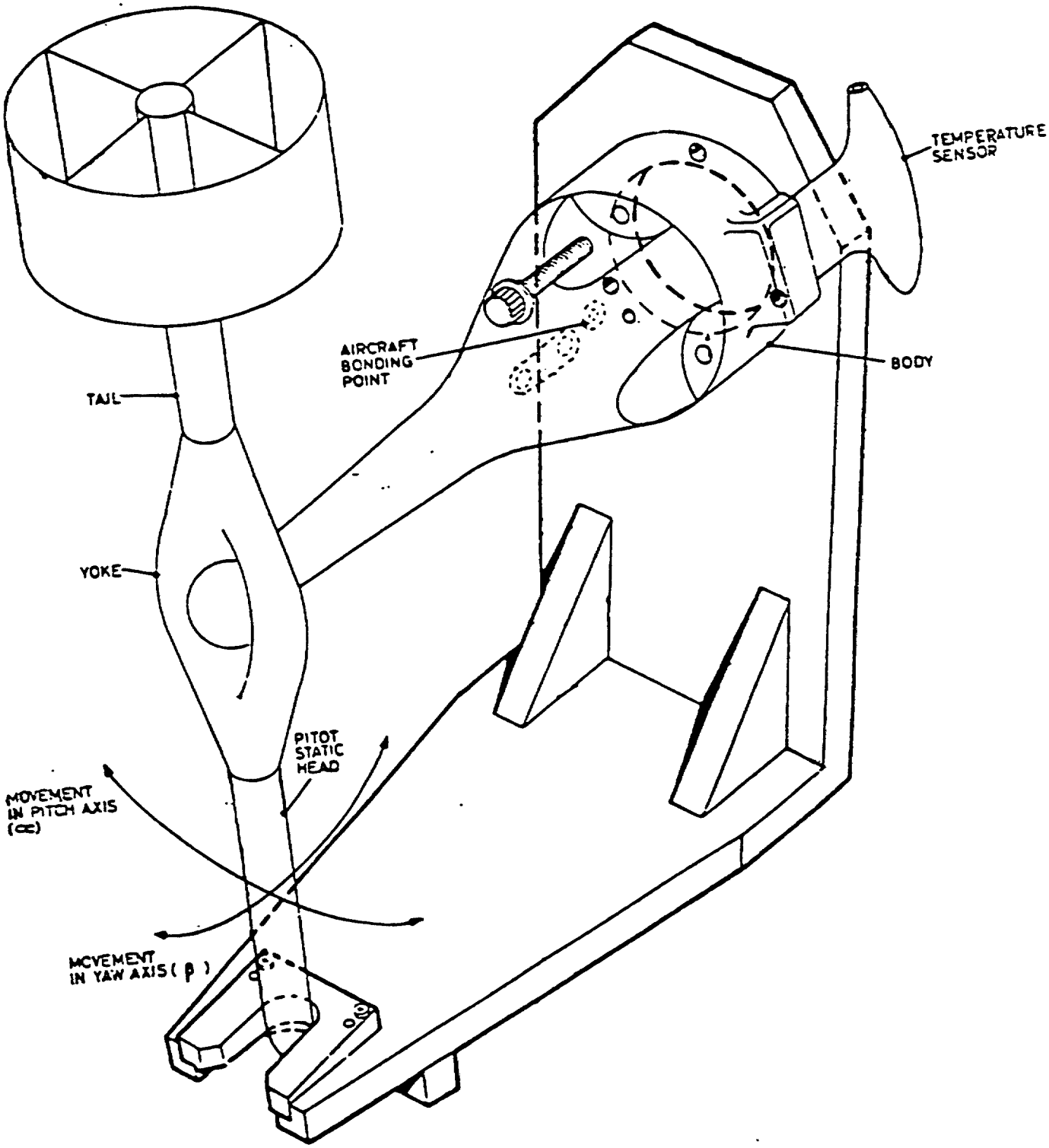


FIGURE 3. Fixture, AADS (Checking)

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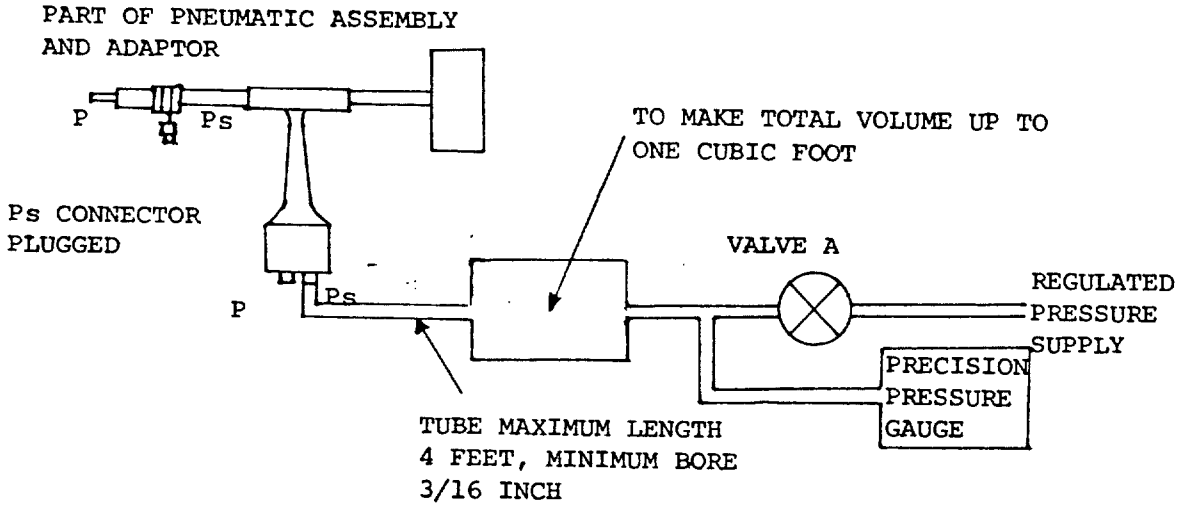


FIGURE 4. Pneumatic circuit - static pressure leak test and static pressure-path flow test

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4.6.2.4 Static pressure-path flow test. With the AADS connected as shown in Figure 4, raise the pressure to simulate an airspeed of 255 knots, and isolate the supply by closing valve A. Remove the plug from the Ps connection of the AADS pneumatic test connector. At an airspeed of 250 knots, start the stopwatch. After 45 secs, the airspeed shall not exceed 50 knots.

4.6.2.5 Total pressure leak test. With the AADS connected as shown in Figure 5, raise the pressure to simulate an airspeed of 255 knots, and isolate the supply by closing valve A. When the airspeed has fallen to 250 knots, start the stopwatch. After 180 secs, the airspeed shall not be less than 175 knots.

4.6.2.6 Total pressure-path flow test. With the AADS connected as shown in Figure 5, raise the pressure to simulate an airspeed of 255 knots, and isolate the supply by closing valve A. Remove the plug from the P_T connection on AADS pneumatic test connector. When the airspeed has fallen to 250 knots, start the stopwatch. After 80 secs, the airspeed shall not exceed 50 knots.

4.6.2.7 Bonding resistance. The resistance measured between the pitot-static head and the aircraft bonding point shall be less than 0.2 ohms. (See Figure 3 for position of bonding point).

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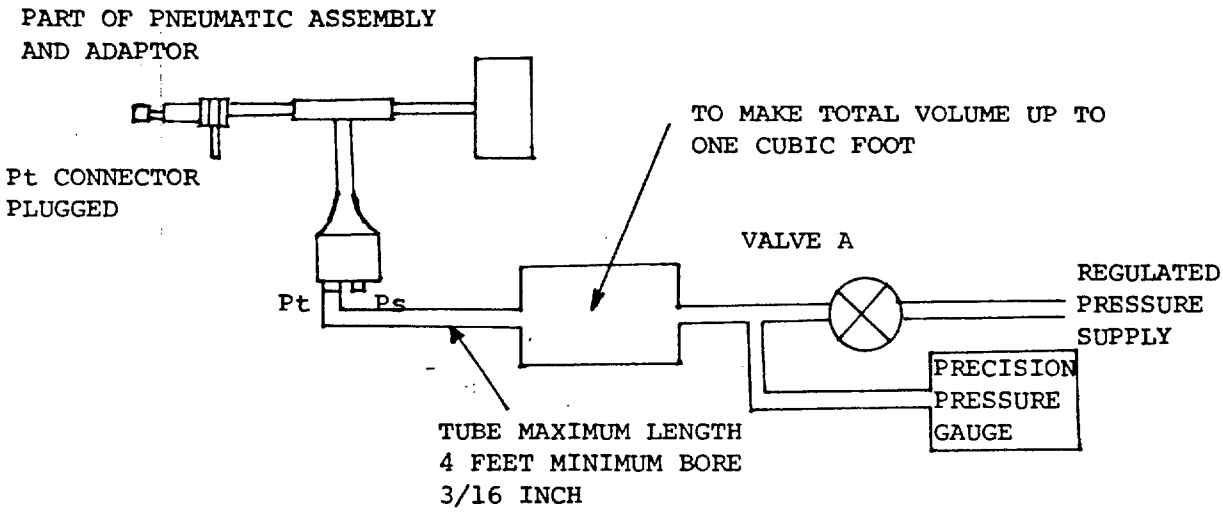


FIGURE 5. Pneumatic circuit - total pressure leak test and total pressure-path flow test

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4.6.2.8 Insulation resistance. The insulation tester shall be used to measure the resistance between the bonding point and the connector pins 12 and 9 joined together.

The insulation resistance at 250V dc shall be not less than 20 M ohms 5 seconds after application.

Care must be taken to avoid accidentally applying 250V dc to other pins, which would result in damage.

4.6.2.9 Air temperature sensor resistance test. Connect the AADS to the AVIM Test Set in accordance with Figure 6.

Connect the DVM across the test points below S12, and select resistance function.

Set S12 to position TEMP and record output. The resistance shall be within the limits defined in Table VI at the ambient temperature. The ambient temperature must be measured to within $\pm$ 0.5°C.

4.6.2.10 Angular outputs accuracy test. Mount the AADS in the checking fixture shown in Figure 3, and connect in accordance with Figure 6. Set S21 UP. Connect the DVM across the test points below S12 and select function VAC.

Set S12 to positions sinαp, cosαp, sin (αp-βp) and cos (αp - βp) in turn, and record each output.

Check the value of αp by marking the intersection of Sinαp and Cosαp readings. The intersection shall fall within the pass band.

Check the value of (αp-βp) by marking the intersectionof Sin (αp-βp) and Cos (αp-βp) readings. The intersection shall fall within the pass band.

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TABLE VI. Air temperature sensor resistance

C Ambient Temperature	Nominal Resistance	Tol.	C Ambient Tolerance	Nominal Resistance	Tol.
10.0	496.0 ohms	±4 ohms	20.5	515.8 ohms	±4 ohms
10.5	497.0 ohms	±4 ohms	21.0	516.7 ohms	±4 ohms
11.0	497.9 ohms	±4 ohms	21.5	517.7 ohms	±4 ohms
11.5	498.9 ohms	±4 ohms	22.0	518.6 ohms	±4 ohms
12.0	499.8 ohms	±4 ohms	22.5	519.5 ohms	±4 ohms
12.5	500.7 ohms	±4 ohms	23.0	520.5 ohms	±4 ohms
13.0	501.7 ohms	±4 ohms	23.5	521.4 ohms	±4 ohms
13.5	502.6 ohms	±4 ohms	24.0	522.3 ohms	±4 ohms
14.0	503.6 ohms	±4 ohms	24.5	523.3 ohms	±4 ohms
14.5	504.5 ohms	±4 ohms	25.0	524.2 ohms	±4 ohms
15.0	505.4 ohms	±4 ohms	25.5	525.2 ohms	±4 ohms
15.5	506.4 ohms	±4 ohms	26.0	526.1 ohms	±4 ohms
16.0	507.3 ohms	±4 ohms	26.5	527.0 ohms	±4 ohms
16.5	508.3 ohms	±4 ohms	27.0	528.0 ohms	±4 ohms
17.0	509.2 ohms	±4 ohms	27.5	528.9 ohms	±4 ohms
17.5	510.1 ohms	±4 ohms	28.0	529.8 ohms	±4 ohms
18.0	511.1 ohms	±4 ohms	28.5	530.8 ohms	±4 ohms
18.5	512.0 ohms	±4 ohms	29.0	531.7 ohms	±4 ohms
19.0	513.0 ohms	±4 ohms	29.5	532.7 ohms	±4 ohms
19.5	513.9 ohms	±4 ohms	30.0	533.6 ohms	±4 ohms
20.0	514.8 ohms	±4 ohms			

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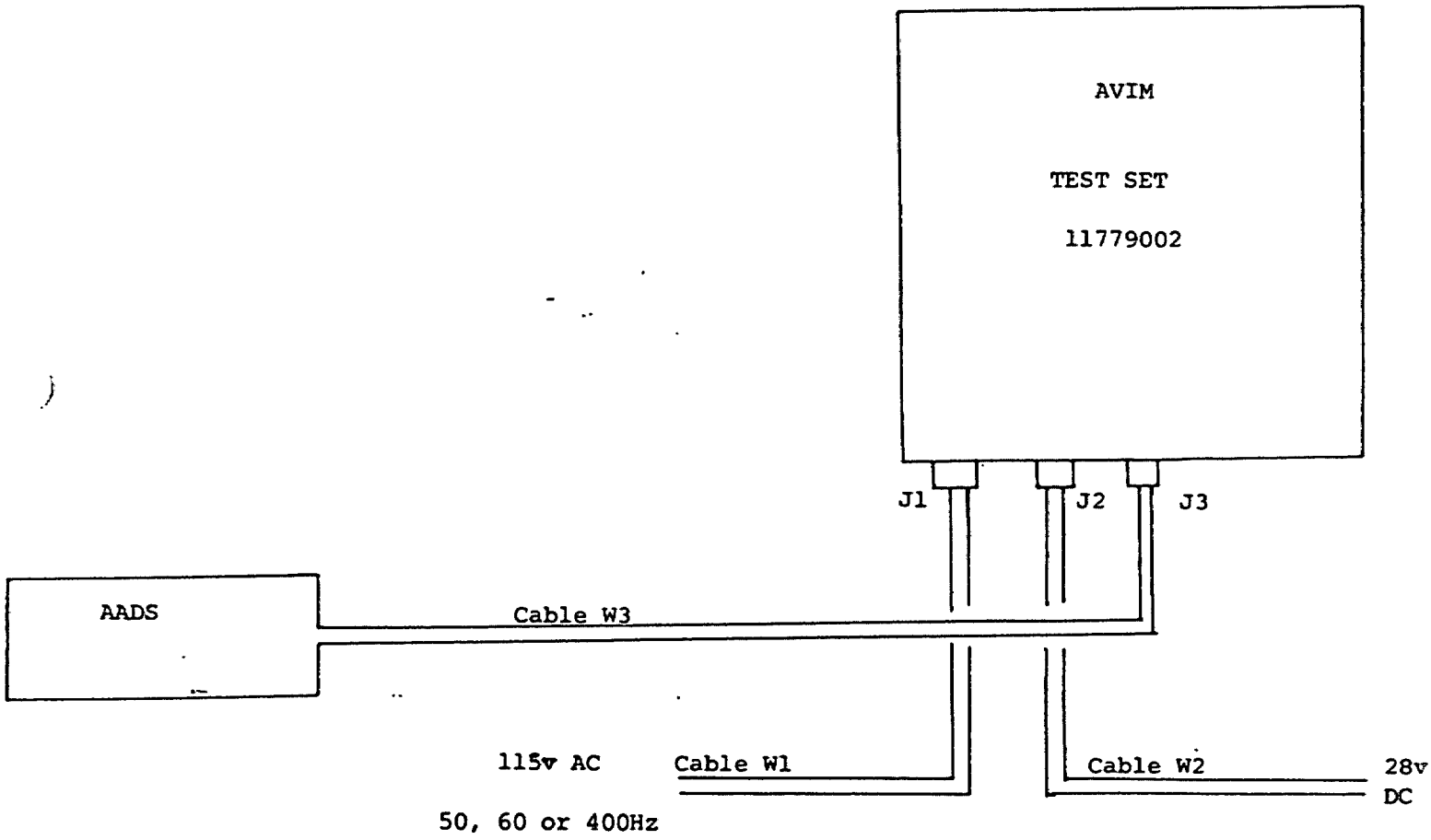


FIGURE 6. Electrical interconnections for test

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4.6.2.10.1 With S12 set to Sinαp, free the pitot-static head from the clamp and move it towards the body (in angle pp) until the stop is reached. Rotate the yoke in p (roll around the body) until the value of Sinαp is less than .05 Volts. Holding this position, switch S12 to Sin (αp-βp) and Cos (αp-βp) and record the readings. Check the value of (αp-βp) by marking the intersection of Sin (αp-βp) and Cos (αp-βp) and record the readings. The intersection shall fall within the pass band.

4.6.2.10.2 Repeat the procedure of paragraph 4.6.2.10.1 but holding the tail end against the stop instead of the head end. Check the value of (αp-βp).

Check that the yoke can be rotated freely through 2 turns in angle p.

4.6.2.11 Heat continuity test. Set S12 to position HTR CONT, and select DVM RESISTANCE function. The resistance shall be 87 ohms $\pm$ 10 ohms.

4.6.3 Physical characteristics.

4.6.3.1 Dimensions. Use standard practices to ensure the requirements of drawing are met.

4.6.3.1.2 Weight. Use item 11 of Table V to ensure that requirements of 3.4.3.2 are met.

4.6.3.1.3 Preconditioning (burn-in). Subject each AADS to the burn in of Appendix B.

4.6.4 Environmental tests.

4.6.4.1 Natural environment. Subject the AADS to the following environments.

4.6.4.2 Altitude. The AADS shall be subjected to the tests of Method 504.1 of MIL-STD-810 for Category 4 equipment. The AADS, nonoperating, shall be subjected to Method 500.1, Procedure I, with pressures simulating air transport at 50,000 feet altitude at a pressure of 2.85 inches of mercury (Hg) for 1 hour. At the conclusions of this test, subject the AADS to the performance tests of Table III.

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4.6.4.3 Temperature. The AADS shall be subjected to test Method 501.1, Procedure II, of MIL-STD-810, except Step 4 temperature shall be 71°C. The system shall then be subjected to the test of Method 502.1, Procedure I, except maximum low temperature shall be -54°C. At the conclusion of this test, subject the AADS to the performance tests of Table III.

4.6.4.4 Humidity., The AADS shall be subjected to test Method 507.1, Procedure II, of MIL-STD-810. In Step 6, measurements shall be taken during the last five hours of the last test cycle. At the conclusion of this test, subject the AADS to the performance tests of Table III.

4.6.4.5 Sand and dust. The AADS shall be subjected to test Method 510.1, Procedure I, of MIL-STD-810, except that the air velocity in Steps 1, 2, and 3 shall be 200 + 100 FPM. Step 3 may proceed immediately after stabilization in Step 2, if desired. At the conclusion of this test, subject the AADS to the performance tests of Table III.

4.6.4.6 Salt atmosphere. The AADS shall be subjected to the test Method 509.1 of MIL-STD-810. Operation of the AADS shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

4.6.4.7 Fungus. The AADS shall be subjected to the test of Method 508.1, Procedure I, of MIL-STD-810. Gasket sealed or other non-hermetic sealed test items shall be opened during test exposure and all internal and external surfaces shall be sprayed with spore suspension. Operation of the equipment is not required, but examination of the concentration after exposure period shall reveal no evidence of viable fungus on any surfaces.

4.6.4.8 Explosive atmosphere. The AADS shall be subjected to test Method 511.1, Procedure I, of MIL-STD-810, except test shall be limited to a maximum altitude of 20,000 feet. At the conclusion of this test, subject the system to the performance tests of Table III.

4.6.4.9 Rain. The AADS shall be subjected to test Method 506.1, Procedure I, of MIL-STD-810. During each thirty minute rain period, the wind shall be maintained for 15 minutes, commencing 5 minutes

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after the start of the period as specified in the procedures. The system shall be operated during the last 10 minutes of the thirty minute period. At the conclusion of this test, subject the AADS to the performace tests of Table III.

4.6.4.10 Service environment.

4.6.4.10.1 Bench handling. The system shall be subjected to Method 516.2, Procedure V, of MIL-STD-810. At the conclusion of this test, subject the AADS to the performance tests of Table III.

4.6.4.11 Induced environment.

4.6.4.11.1 Vibration. The AADS shall be subjected to test Method 514.2, Procedure I, Part I, Table 514.2-III, of MIL-STD-810, except that the frequency shall be limited to 500 cycles (reference Figure 2). At the conclusion of this test, subject the AADS to the performance test of Table III.

4.6.4.11.2 Shock. The AADS shall be subjected to:

a. Method 516.2 of MIL-STD-810, Procedure I, Figure 516.2-2 for flight vehicle equipment. The test "g" value shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter. The pulse test level and duration are defined in Paragraph 3.2.24.6.1 of MIL-E-5400.

b. Method 516.2, Procedure III for mounting base crash safety. The pulse test level and duration are defined in Paragraph 3.2.24.6.2 of MIL-E-5400.

At the conclusion of this test, subject the AADS to the performance test of Table III.

4.6.4.11.3 Transportability. The system shall be subjected to the test of Method 514.2, Procedure X, of MIL-STD-810. At the conclusion of this test, subject the AADS to the performance tests of Table III.

4.6.4.12 Electromagnetic radiation. The AADS shall be installed in laboratory test fixtures in a manner that will simulate actual service usage and tested to MIL-STD-461 for the requirements of 3.4.8.

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5. PACKAGING.

5.1 Preservation, packing, packaging and marking. The AADS shall be preserved, packaged, packed, and marked in accordance with the requirements of the preservation and packaging data sheet P11776800 at the level of protection specified in the contract (see 6.2b).

5.2 Commercial packaging shall be in accordance with ASTM-D3951.

6. NOTES.

6.1 Intended use. This equipment is intended for use in the AH-1S (modernized) helicopters.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging Data Sheet (see 5.1).

c. Selection of applicable levels of preservation, packaging, and packing.

d. Applicable stock number.

e. First article inspection requirements.

6.3 Previous revisions of MIL-D-70487 were for the US Army Armament Research and Development Center internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1270-A055)

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APPENDIX A

Signal Interface Requirements Ident Number 1

Power Signal Interface Requirements	
Nomenclature	AADS REF
Type and Frequency	AC 400± 20Hz
Conductor Type and Gauge	Twisted Shielded Pair 24 AWG
Origin	EPU
Destination	AADS
Destination Impedence	>300 ohms
Source	EPU
Source Impendence	<300 ohms
Power Required	180 mW
Phase	N/A
Maximum Current	25mA
Nominal Voltage	18.45V Peak to Peak
Maximum Voltage	20.5V Peak to Peak
Minimum Voltage	12.0V Peak to Peak
Power Factor	N/A
Regulation	N/A

REMARKS

Excitation Signal for AADS Angle Resolvers

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APPENDIX A

Signal Interface Requirements — Ident Number 2

Analog Signal Interface Requirements	
Nomenclature	SIN Alpha
Type	AC Analog 400 ± 20Hz
Conductor Type and Gauge	Twisted Shielded Triple [1] 24 AWG
Origin	AADS
Destination	EPU
Destination Impendence	>10K ohms
Source	AADS
Source Impendence	<300 ohms
Voltage Range	0.454 V_R SIN Alpha Peak to Peak [2]
Scale Factor	0.454 V_R Cos Alpha
Parameter Range	-1 to +1
Parameter Accuracy	Equivalent to an angle accuracy of ±2 degrees
Phasing	N/A
RMS Null	0 VDC when Sin Alpha = 0
Phase shift	17 ± 3 degrees lead relative to AADS Ref.
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	9.6 V Peak to Peak

REMARKS

[1] Sin Alpha, Cos Alpha and Alpha RTN

[2] V_R = AADS Ref Peak to Peak Voltage

Angular position signal from AADS Resolver

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APPENDIX A

Signal Interface Requirements — Ident Number 3

Analog Signal Interface Requirements	
Nomenclature	Cos Alpha
Type	AC Analog 400 ± 20 Hz
Conductor Type and Gauge	Twisted Shielded Triple [1] 24 AWG
Origin	AADS
Destination	EPU
Destination Impendence	>10K ohms
Source	AADS
Source Impendence	<300 ohms
Voltage Range	0.454 V_R Volts Peak to Peak[2]
Scale Factor	0.454 V_R Cos Alpha
Parameter Range	+1 to -1
Parameter Accuracy	Equivalent to an angle accuracy of ± 2 degrees
Phasing	N/A
RMS Null	0 VDC when Cos Alpha = 0
Phase Shift	17 ± 3 degrees lead relative to AADS Ref.
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	9.6 V Peak

REMARKS

[1] Sin Alpha, Cos Alpha and Alpha RTN

[2] V_R = AADS Ref Peak to Peak Voltage

Angular position Signal from AADS Resolver

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APPENDIX A

Signal Interface Requirements — Ident Number 4

Analog Signal Interface Requirements	
Nomenclature	Sin (Alpha-Beta)
Type	AC Analog 400± 20Hz
Conductor Type and Gauge	Twisted Shielded Triple [1] 24 AWG
Origin	AADS
Destination	EPU
Destination Impendence	>10K ohms
Source	AADS
Source Impendence	<300 ohms
Voltage Range	0.454 V_R Volts Peak to Peak [2]
Scale Factor	0.454 V_R Sin (Alpha-Beta)
Parameter Range	+1 to -1
Parameter Accuracy	Equivalent to an angle accuracy of ±2 degrees
Phasing	N/A
RMS Null	0 VDC when Sin (Alpha-Beta) = 0
Phase Shift	17 ±3 degrees lead relative to AADS Ref.
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	9.6V Peak

REMARKS

[1] Sin (Alpha-Beta), Cos (Alpha-Beta) and Alpha-Beta RTN

[2] V_R = AADS REF Peak to Peak Voltage

Angular position signal from AADS Resolver

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APPENDIX A

Signal Interface Requirements — Ident Number 5

Analog Signal Interface Requirements	
Nomenclature	Cos (Alpha-Beta)
Type	AC Analog 400 ± 20Hz
Conductor Type and Gauge	Twisted Shielded Triple[1] 24 AWG
Origin	AADS
Destination	EPU
Destination Impendence	>10K ohms
Source	AADS
Source Impendence	<300 ohms
Voltage Range	0.454 V_R Vols Peak to Peak [2]
Scale Factor	0.454 V_R Cos (Alpha-Beta)
Parameter Range	+1 to -1
Parameter Accuracy	Equivalent to an angle accuracy of ± 2 degrees
Phasing	N/A
RMS Null	0 VDC when Cos (Alpha-Beta) = 0
Phase Shift	17 ±3 degrees lead relative to AADS Ref.
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	9.6 V Peak

REMARKS

[1] Sin (Alpha-Beta), Cos (Alpha-Beta) and Alpha-Beta RTN

[2] V_R = AADS Ref Peak to Peak Voltage

Angular position signal from AADS Resolver.

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APPENDIX A

Signal Interface Requirements — Ident Number 6

Analog Signal Interface Requirements	
Nomenclature	Air Temperature (ATS)
Type	Resistance
Conductor Type and Gauge	Twisted Shielded Pair 24 AWG
Origin	AADS
Destination	EPU
Destination Impendence	N/A
Source	AADS
Source Impendence	477.17 ohms at $0^{\circ}C$
Voltage Range	N/A
Scale Factor	Approx 2 ohms/$^{\circ}C$ Rt = 477.17 (1+.0039 58t - .000000583t^2) where Rt resistance in ohms at $t^{\circ}C$.
Parameter Range	374.4 to 609.9 ohms
Parameter Accuracy	$\pm$0.6 ohms at $0^{\circ}C$
Phasing	N/A
RMS Null	N/A
Phase Shift	N/A
Sensitivity	N/A
Maximum Current	5 mA
Maximum Voltage	3.05V

REMARKS

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APPENDIX B

BURN-IN REQUIREMENTS

Burn-in is to be applied to all deliverable systems or LRU's.

Test Environments

Environmental exposure for the AADS during burn-in is Level F of MIL-STD-781B. Cycle time shall be determined by thermal survey. Chamber temperature change shall be programmed equal to or greater than 5° per minute. Figure 1 illustrates one cycle of the environmental profile described below.

Temperature	-50°C to +71°C (-65°F to 160°F)
Temperature Cycling	Temperature cycling shall be time to stabilize at low temperature followed by time to stabilize at the high temperature, plus 2 hours.
Vibration	2.2G ± 10% peak acceleration value at 30 ± 5 Hz measured at the mounting points. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.
Equipment On-Off Cycling	Equipment off during cooling cycle and on during heating cycle.
Input Voltage	Specified voltage +5 -2%.
Input Voltage Cycling	a. Nominal voltage for first 1/2 of "ON" time starting at the beginning of the heating period. b. Momentary off. c. 100% of nominal for next 1/3 of "ON" time. d. Monemtary off. e. 90% of nominal for last 1/3 of "ON" time.

NOTE: Voltage Cycling will occur on the primary AADS power (28 VDC) only.

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APPENDIX B

Number of Cycles

The minimum number of cycles shall be five (5) with the last two (2) cycles failure free.

Duty Cycle Information

During equipment ON time, mode selection time shall be apportioned as follows:

Test	95% ± 5%

During operation, the display and BIT Indicators shall be monitored.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-70487 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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